

LUND UNIVERSITY

Subspaces of the Hardy space

Operator and function theory

Bergman, Alex

2026

Link to publication

Citation for published version (APA):
Bergman, A. (2026). *Subspaces of the Hardy space: Operator and function theory*. MediaTryck Lund.

Total number of authors:
1

General rights
Unless other specific re-use rights are stated the following general rights apply:
Copyright and moral rights for the publications made accessible in the public portal are retained by the authors and/or other copyright owners and it is a condition of accessing publications that users recognise and abide by the legal requirements associated with these rights.
• Users may download and print one copy of any publication from the public portal for the purpose of private study or research.
• You may not further distribute the material or use it for any profit-making activity or commercial gain
• You may freely distribute the URL identifying the publication in the public portal

Read more about Creative commons licenses: https://creativecommons.org/licenses/

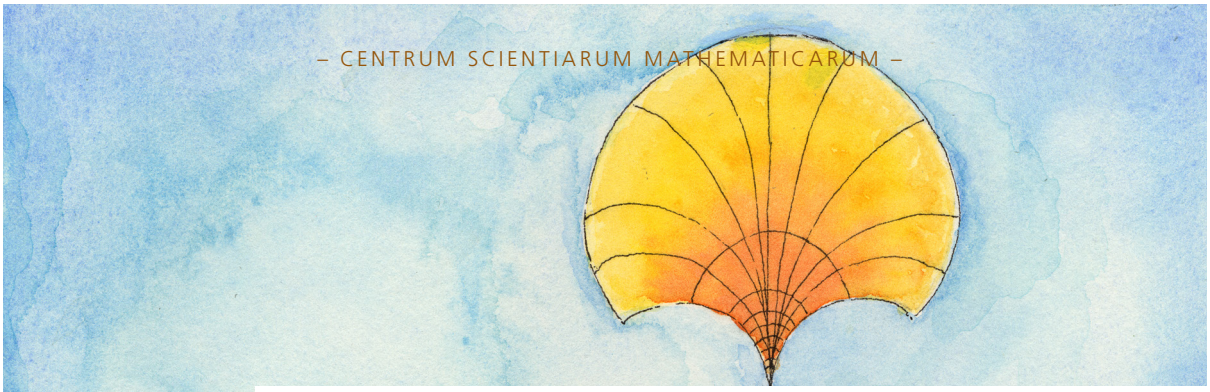
Take down policy
If you believe that this document breaches copyright please contact us providing details, and we will remove access to the work immediately and investigate your claim.

LUND UNIVERSITY

PO Box 117
221 00 Lund
+46 46-222 00 00

Download date: 29. Oct. 2026

– CENTRUM SCIENTIARUM MATHEMATICARUM –



Subspaces of the Hardy space

Operator and function theory

ALEX BERGMAN



Lund University
Faculty of Science
Centre for Mathematical Sciences
Mathematics

Subspaces of the Hardy space
Operator and function theory

Subspaces of the Hardy space
Operator and function theory

by Alex Bergman



LUND
UNIVERSITY

Thesis for the degree of Doctor of Philosophy
Thesis advisors: Prof. Alexandru Aleman
Faculty opponent: Dr. Ole Brevig

To be presented, with the permission of the Faculty of Science of Lund University, for public criticism at Matematikcentrum (Hörmandersalen) on friday, the 5th of June 2026 at 13:15.

Organization	Document name
LUND UNIVERSITY Department of Mathematics Box 124 SE–221 00 LUND Sweden	**DOCTORAL THESIS** Date of disputation 2026-06-05 Sponsoring organization
Author(s) Alex Bergman	

Title and subtitle
Subspaces of the Hardy space Operator and function theory

Abstract
This thesis investigates subspaces of the Hardy space, especially ones defined via analytic continuation, and their applications to operator theory. The main function theoretic properties investigated are the distribution of zeros and extremal functions. The operator theory investigations are centered around the concept of invariant subspaces and cyclic vectors.

Key words
Hardy space, operator theory, analytic functions

Classification system and/or index terms (if any)

Supplementary bibliographical information	Language
	English
ISSN and key title 1404-0034	ISBN 978-91-8104-931-2 (print) 978-91-8104-932-9 (pdf)
Recipient's notes	Number of pages 202 / Price
	Security classification

I, the undersigned, being the copyright owner of the abstract of the above-mentioned dissertation, hereby grant to all reference sources the permission to publish and disseminate the abstract of the above-mentioned dissertation.

Signature _____ Date 2026-04-15

DOKUMENTDATABLAD enl SIS 61 41 21

Subspaces of the Hardy space

Operator and function theory

by Alex Bergman



LUND
UNIVERSITY

Cover illustration: Isa Clementsson.

© Alex Bergman 2026

Faculty of Science, Department of Mathematics

Doctoral Theses in Mathematical Sciences 2026:4
ISSN: 1404-0034
ISBN : 978-91-8104-931-2 (print)
ISBN : 978-91-8104-932-9 (pdf)
LUNFMA-1050-2026

Printed in Sweden by Media-Tryck, Lund University, Lund 2026

Printed matter
3041 0903

Media-Tryck is a Nordic Swan Ecolabel
certified provider of printed material.
Read more about our environmental
work at www.mediatryck.lu.se

MADE IN SWEDEN

Contents

Subspaces of the Hardy space Operator and function theory 1

1 The model theory of operators 3

Scientific publications 21

Paper I: On cyclicity in de Branges-Rovnyak spaces 23

Paper II: Chebyshev polynomials corresponding to a vanishing weight 49

Paper III: A sharp entropy condition for the density of angular derivatives 83

Paper IV: Hörmander's inequality and point evaluations in de Branges space 91

Paper V: Invariant Subspaces for Generalized Differentiation and Volterra Operators 119

Paper VI: Inner approximations of doubling weights with applications to Beurling-Malliavin theory in Toeplitz kernels 167

List of publications

This thesis is based on the following publications, referred to by their Roman numerals:

I **On cyclicity in de Branges-Rovnyak spaces**

A. Bergman
Indiana University Mathematics Journal, 2024, 73(4), pp. 1307–1329

II **Chebyshev polynomials corresponding to a vanishing weight**

A. Bergman, O. Rubin
Journal of Approximation Theory, 2024, 301, 106048, 25

III **A sharp entropy condition for the density of angular derivatives**

A. Bergman
Comptes Rendus Mathématique. Académie des Sciences. Paris, 2025, 363, pp. 29–33

IV **Hörmander's inequality and point evaluations in de Branges space**

A. Bergman
Revista Matemática Iberoamericana, 2026, no. 2, pp. 551–572

V **Invariant Subspaces for Generalized Differentiation and Volterra Operators**

A. Aleman, A. Bergman
Preprint, 2025

VI **Inner approximations of doubling weights with applications to Beurling-Malliavin theory in Toeplitz kernels**
A. Bergman
Preprint, 2025

All papers are reproduced with permission of their respective publishers.

Acknowledgements

I would like to thank everyone that has made the past five years of my life a great experience. I believe that when I look back on these years in the future I will remember them as some of the best of my life. Unfortunately, there are quite a few too many to mention by name and any omission below is unintentional.

I will begin by thanking Alexandru for being a great mentor and friend. Your encouragement and honest feedback has been very important to me. I would also like to thank you for introducing me to a fascinating area of research.

I would also like to thank my co-supervisors Yacin, for teaching me harmonic measure, and Adem, for reading all of my manuscripts, providing useful feedback, and for all of our longs chats throughout the years. I would also like to thank Eskil for giving me, much needed, feedback when I was still a student.

To my coauthor and friend from my first year in Lund, Olof, I would like to say thank you for all of the time we spent together, both mathematical and otherwise. I have some very fond memories of picking berries and mushrooms together and doing math at your parents' summer place in Sankt Olof.

I would like to thank Jorge, Joakim and Thanos for all the training, skiing, evenings hanging out together and general good times.

To Frej, Joakim, Raul, Thanos and Olof for all the fun we had together and the conferences we went to. I would also like to thank you for being academic older brothers to me, especially Frej.

Stefano, my roommate for a year. I really enjoyed living together.

To the rest of the doctoral student group German, Mårten, Mats, Magnus, Alejandro, Wilhelm, Ena, Valentina, Anna, Erik, Jörg, Samuele, Giang, Tom, Marvin and Jan. Thank you for all the great times.

The senior members of the department Erik, Anitha, Evgeniy, Sandra, Anna-Maria also deserve a thank you for all their hard work making the department a pleasant place to be.

Jag vill tacka mina vänner Isak, Ludvig, Lukas och Rickard som jag haft sedan barndomen i Umeå. Ni har alltid gjort det extra roligt att åka hem under julen och sommaren.

Till sist vill jag nämna min familj Torbjörn, Cindy, Jonathan och Isa. Tack för all kärlek och stöttning jag fått genom åren. Ni har betytt allt för mig.

Populärvetenskaplig sammanfattning på svenska

Alla funktioner vi är vana vid från grundskolan och gymansiet är analytiska. Exempel är polynom, exponentialfunktionen, de trignometriskafunktionerna som sinus, kosinus och deras inversa funtkioner. Studiet av analytiskafunktioner heter funktionsteori och är numera ett klassiskt och centralt ämne inom matematiken med ett brett inflytande inom geometri, talteori, differentialekvationer, topologi, med flera. Infinitesimalkalkylen upptäcktes under 1600-talet av Isaac Newton och Gottfried Leibniz och funktionsteorin tog fart århundradet efter det.

En operator är ett objekt som beskriver en fysikalisk lag. Ett typsikt problem kan vara att man vill veta temperaturen i mitten eller botten av en sjö men att det är svårt att ta sig dit. Istället för det kan man mäta temperaturen vid stranden och använda att temperaturen är en lösning av en ekvation som beskrivs av en viss operator för att räkna ut temperaturen i mitten.

I denna avhandling utforskar vi vad som händer när man studerar funktionsteori med hjälp av operatorteori (studiet av operatorer) eller vice-versa. Att detta är möjligt och givande är egentligen inte förvånande och har varit känt sedan operatorteorins uppkomst i början av 1900-talet. Trots detta finns det nya metoder och spännande problem kvar att lösa. I detta verk behandlar vi en del av dessa.

Subspaces of the Hardy space
Operator and function theory

Chapter 1

The model theory of operators

1.1 Introduction

The basic idea in the model theory of operators is the following. One starts with a collection of operators each defined on a vector space (which may depend on the operator). New vector spaces and a collection of bijective linear transformations from the first vector spaces to the second vector spaces are defined. Next one identifies the image of the operator under the bijective transformations. These are the model operators.

The program as formulated makes little sense since the operators could have nothing to do with each other. In practice one chooses operators, transformations, and vector spaces that share certain common features. The goal is typically to transform the given operators to certain canonical operators with simpler algebraic structure. Instead of trying to give an overly formal definition, we present some illustrative examples.

Example 1. The operators considered are bounded self-adjoint operators on a Hilbert space with one-dimensional spectrum. The model operators are the operators of multiplication by independent variable on $L^2(\mathbb{R}, \mu)$. The bijective linear transformations are the unitary operators diagonalizing a given self-adjoint operator given by the spectral theorem.

Example 2. The Hilbert space is $\ell^2(\mathbb{Z}_+)$. The operator is the shift operator S. The model operator is multiplication by the independent variable z on the Hilbert space H^2 consisting of analytic power series in the unit disc with square summable coefficients. The bijective transformation is the map $U : \{a_n\}_{n=0}^{\infty} \mapsto \sum a_n z^n$.

Example 3. The operators are constant coefficient linear PDEs,

$$L = \sum_{\alpha} a_{\alpha} D^{\alpha}.$$

The bijective linear operator is the Fourier transform. The model operators are multiplication by polynomials.

Example 4. The operators are contractions on a Hilbert space and the model operators are compressions of the shift operator onto the $*$-invariant subspaces of the Hardy space. This is the model theory of Sz.-Nagy and Foiaș.

Once the program has been formulated it is possible to work on one aspect without direct mention of the others. In particular, it is often the case that the space on which the model operator acts has a complicated or indirect definition. This is not too surprising since one has simplified the algebra involved and thus must pay a price. Of the articles below only paper I and paper V deal with operators directly and only in paper V is the model explicitly constructed.

In this thesis the vector spaces which the model operators act on will be spaces of holomorphic functions. In this way, the typical questions of operator theory, such as understanding the spectrum, cyclic vectors, invariant subspaces, and completeness of eigenfunctions, will be transformed into problems on the existence or non-existence of holomorphic functions with certain prescribed values, for example, zero sets and interpolation. These problems are the typical bread and butter of complex analysis and if one is satisfied with merely holomorphic solutions then these questions can be resolved using the classical theorems of Weierstrass and Mittag-Leffler. However, the requirement that the function should also be an element of an appropriate vector space causes serious problems and must be dealt with on a case by case basis. This motivates a detailed study of the spaces involved.

In this thesis the best demonstration of this program is paper V which carries it out in full. A class of operators is abstractly defined and model operators are constructed for them. Then deep theorems on the holomorphic function spaces arising in the construction of the model operators are used to describe the invariant subspaces of the operators.

Before we begin our analysis we wish to motivate the spaces of functions considered and in particular the simple geometry of the domain. To do this let us consider the Hilbert space case. Since all (separable) Hilbert spaces are unitarily equivalent there is a lot of freedom when choosing the vector space and our goal is to look for the

simplest possible representation. The uniformization theorem tells us that there are only three possible choices for a simply connected Riemann surface. The sphere, S^2, the unit disc, $\mathbb{D}$, and the complex plane $\mathbb{C}$. The only holomorphic maps from the sphere to the unit disc or plane are constants and the holomorphic maps from the sphere to itself are rational functions. Although, the case of rational and, in particular, polynomial mappings is also interesting we shall not consider it seriously in this treatise. Instead we focus on the case of the unit disc and plane which offer a greater variety of possibilities.

1.2 The Hardy space

The vector space which we shall be principally concerned with is the Hardy space, H^p, which we introduce, for all $0 < p \leq \infty$ in both the unit disc,

$$\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$$

and the upper half-plane,

$$\mathbb{C}_+ = \{z \in \mathbb{C} : \mathrm{Im}(z) > 0\}\,.$$

In the unit disc we let $H^p(\mathbb{D})$ consist of all holomorphic functions, such that the norm (in the case $1 \leq p < \infty$)

$$\|f\|_p^p = \sup_{0<r<1} \frac{1}{2\pi} \int_0^{2\pi} |f(re^{i\theta})|^p d\theta < \infty. \tag{1.1}$$

In the upper half-plane condition (1.1) is replaced by the condition

$$\|f\|_p^p = \sup_{y>0} \frac{1}{2\pi} \int_{\mathbb{R}} |f(x+iy)|^p dx < \infty. \tag{1.2}$$

For $p = \infty$ the Hardy space is defined to be the vector space of bounded holomorphic functions on the disc or the upper half-plane, as the case may be. General references for the theory of Hardy spaces include [10, 13]. Let us remark that, except in the case $p = \infty$, the Hardy space of the upper half-plane is not the pullback of the Hardy space of the unit disc under a conformal map $\mathbb{C}_+ \to \mathbb{D}$.

A fundamental result is that Hardy space functions have radial boundary values at almost every point on the unit circle $\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$ or along lines parallel to the imaginary axis approaching $\mathbb{R}$ corresponding to the disc and half-plane. The

boundary function, which we denote by the same letter, belongs to L^p and for $p \geq 1$ the function may be recovered from its boundary function via the Poisson integral formula, which in the disc takes the form

$$f(z) = \int_{\mathbb{T}} \frac{1-|z|^2}{|\zeta - z|^2} f(\zeta) dm,$$

where dm is the normalized arc-length measure on the unit circle. In addition, non-trivial Hardy space functions cannot vanish on subsets of the boundary of positive measure. A quantitative version of this statement is that the logarithmic integral is finite, that is

$$\int_{\mathbb{T}} \log|f(\zeta)| dm > -\infty.$$

Functions in the Hardy space enjoy Nevanlinna factorization

$$f(z) = \lambda S_\mu(z) B(z) F(z),$$

where λ is a unimodular constant, F is an outer function, B is a Blaschke product, and S_μ is a singular inner function. The product $\theta = BS_\mu$ is called the inner factor of f. The three factors have explicit formulas. The outer function is given by

$$F(z) = \exp\left(\int \frac{\zeta + z}{\zeta - z} \log|f(\zeta)| dm \right).$$

The Blaschke factor is given by

$$B(z) = z^m \prod \frac{-\overline{a_n}}{|a_n|} \frac{z - a_n}{1 - \overline{a_n} z}, \quad \{a_n\} \subset \mathbb{D}, \text{ and } \sum 1 - |a_n| < \infty,$$

where the a_ns are the zeros of f. The singular inner factor is given by

$$S_\mu(z) = \exp\left(- \int \frac{\zeta + z}{\zeta - z} d\mu(\zeta) \right),$$

where μ is a nonnegative singular measure on the circle. Analogous results also hold in the half-plane case. A holomorphic self map of the unit disc is called inner if it has radial boundary values with modulus 1 almost everywhere. The following theorem of Beurling plays a central role.

Theorem 1 (Beurling, [8]). *Let $1 \leq p < \infty$ and $M \subsetneq H^p(\mathbb{D})$ be a nontrivial closed subspace. Suppose M is invariant under multiplication by z. Then there exists an inner function $\theta : \mathbb{D} \to \mathbb{D}$, such that $M = \theta H^p$.*

In this work we shall mainly focus on the orthogonal complement of θH^2,

$$K_\theta = H^2 \ominus \theta H^2.$$

The relevance of K_θ to the model theory of operators comes from the following result due to Sz.-Nagy-Foiaş.

Theorem 2 (Sz.-Nagy-Foiaş, [16], see also Theorem 4.1. in [12]). *Let $T : H \to H$ be a contraction on a Hilbert space. Suppose*

(i) T^n *goes strongly to zero.*

(ii) T *is almost unitary, i.e.*

$$\dim \mathit{Ran}(I - T^*T) = \dim \mathit{Ran}(I - TT^*) = 1.$$

Then there exists an inner function θ, such that T is unitarily equivalent to the compression of the shift operator onto K_θ,

$$T \simeq S_\theta = P_\theta M_z,$$

where P_θ is the orthogonal projection from H^2 onto K_θ and $M_z f = zf$.

The conditions are not essential in the sense that dropping condition (i) merely adds an orthogonal L^2-part and dropping condition (ii) requires one to replace K_θ by a vector-valued version.

In principle we have now reduced the study of Hilbert space contractions to the study of the model operator S_θ, however, the computations involved in the construction of θ are often intractable and to extract any value from the model we must understand the space K_θ. One simple, but important, observation is that the functions

$$k_\lambda(z) = \frac{1 - \overline{\theta(\lambda)}\theta(z)}{1 - \overline{\lambda} z}, \text{ for } \lambda \in \mathbb{D},$$

belong to K_θ. They are called the reproducing kernels and satisfy

$$f(\lambda) = \langle f, k_\lambda \rangle_2, \text{ for } f \in K_\theta.$$

From now on we shall mainly focus on the simplest nontrivial examples which turn out to be surprisingly rich. We describe these in the next section.

1.3 Meromorphic inner functions

As we have seen inner functions may be factorized as a product of a singular inner factor and a Blaschke product,

$$\theta(z) = B(z)S_\mu(z).$$

The spectrum, $\sigma(\theta)$, of an inner function is, by definition, the subset of the unit circle where θ does not extend analytically. It is equal to the union of the accumulation points of the zeros of the Blaschke product and the closed support of the singular measure μ. Thus it is void if and only if θ is a finite Blaschke product. This case is too simple and thus we consider the next simplest case when $\sigma(\theta)$ consists of one point which we fix to be one,

$$\sigma(\theta) = \{1\}.$$

Thus θ is the product of a Blaschke product whose zeros accumulate only at the point one and the singular inner factor with the point measure $\mu = a\delta_1$. It turns out that in this case all functions in K_θ extend to meromorphic functions on the Riemann sphere except the point one. Thus it makes sense to move the point one to infinity and consider meromorphic functions defined on the plane instead. Doing this we are led to meromorphic inner function, i.e. inner function $\Theta : \mathbb{C}_+ \to \mathbb{D}$ which extend to meromorphic functions in the whole complex plane and the model subspaces

$$K_\Theta = H^2(\mathbb{C}_+) \ominus \Theta H^2(\mathbb{C}_+).$$

Meromorphic inner functions have a particularly simple Nevanlinna factorization

$$\Theta(z) = e^{iaz}B(z),\ z \in \mathbb{C}_+,$$

where $a \geq 0$ and B is a Blaschke product whose zeros accumulate only at infinity. Let us recall that the Blaschke condition in the upper half-plane is

$$\sum \frac{\mathrm{Im}(a_n)}{1 + |a_n|^2} < \infty.$$

Probably the best reference for meromorphic inner functions, in particular in connection with problems considered in this thesis is the paper [15].

1.4 De Branges spaces

De Branges spaces are the same as model spaces generated by meromorphic inner functions, but sometimes offer a better perspective. They are defined by a symbol E which is a Hermite-Biehler function, i.e. an entire function satisfying

$$|E(z)| > |E(\overline{z})|, \quad \text{for all } z \in \mathbb{C}_+.$$

The de Branges space is then

$$\mathcal{H}(E) = \left\{ f : f \text{ is entire }, \ f/E, \ f^{\#}/E \in H^2(\mathbb{C}_+) \right\},$$

where $f^{\#}(z) = \overline{f(\overline{z})}$. The norm is given by the L^2 norm of f/E on the real line. The connection to meromorphic inner functions comes from the fact that

$$\Theta_E(z) = E^{\#}(z)/E(z),$$

is a meromorphic inner function and multiplication $E : K_{\Theta_E} \to \mathcal{H}(E)$ is an isometric transformation. Conversely, to every meromorphic inner function Θ there exists a (not unique) Hermite-Biehler function E, such that $\Theta = E^{\#}/E$.

1.5 Summary of research papers

In this section we give a brief summary of the research papers which this thesis is based on. To not overwhelm the reader we present only a selection of the results and the statements of the theorems here are often simpler special cases of the theorems in the respective papers.

Paper I

Paper I [3] deals with the question of describing the cyclic vectors of the shift operator in de Branges-Rovnyak spaces. De Branges-Rovnyak spaces can be defined by specifying their reproducing kernel function. Let $b : \mathbb{D} \to \mathbb{D}$ be a holomorphic self-map of the unit disc. Then the de Branges-Rovnyak space $\mathcal{H}(b)$ is the Hilbert space of analytic functions on the unit disc with kernel

$$k_\lambda(z) = \frac{1 - \overline{b(\lambda)}b(z)}{1 - \overline{\lambda}z}.$$

If b is an inner function we obtain a model space. Model spaces are never shift invariant, however, if the function b satisfies

$$\int \log(1 - |b|)dm > -\infty, \tag{1.3}$$

then $M_z f = zf$ is a bounded operator on $\mathcal{H}(b)$. In fact, this is a necessary condition as well. From now on we assume that b satisfies (1.3). A function $f \in \mathcal{H}(b)$ is called cyclic if the linear span of the orbit

$$\{z^n f : n \geq 0\}$$

is dense in $\mathcal{H}(b)$. We denote the closure of the linear span of the orbit of f by $[f]$. Sarason has proved that polynomials are dense in $\mathcal{H}(b)$ or, equivalently, that the function 1 is a cyclic vector. It turns out that the de Branges-Rovnyak space is contractively contained inside the Hardy space H^2, thus

$$\|1 - pf\|_2 \leq \|1 - pf\|_{\mathcal{H}(b)},$$

for all functions $f \in \mathcal{H}(b)$ and polynomials p. It follows that if f is cyclic for a de Branges-Rovnyak space it is also cyclic in H^2 and hence an outer function. Thus in questions related to cyclicity we may restrict to outer functions. From (1.3) it follows that there exists a unique outer function a with $a(0) > 0$ and

$$|a|^2 + |b|^2 = 1, \text{ almost everywhere.}$$

It turns out that $aH^2 = \{af : f \in H^2\}$ is contained inside $\mathcal{H}(b)$ and one of the basic results of paper I is that if f is outer, then

$$aH^2 \subset [f].$$

It follows that one can reduce the question of cyclicity to the question of the containment of $(aH^2)^\perp$ in $[f]$ (here the orthogonal complement is taken with respect to the $\mathcal{H}(b)$ norm). We are naturally led to consider two cases

$$\dim(aH^2)^\perp < \infty,$$

and

$$\dim(aH^2)^\perp = \infty.$$

The first case is much simpler and a complete description was obtained. For the statement we need the notion of Carathéodory angular derivative.

Definition 1. *Let $b : \mathbb{D} \to \mathbb{D}$ denote a holomorphic self map of the unit disc. The function b is said to have an angular derivative at $\lambda \in \mathbb{T}$ in the sense of Carathéodory if the following two conditions hold*

(i) b has a non-tangential limit at λ and $|b(\lambda)| = 1$.

(ii) The derivative, b', has finite non-tangential limit at λ.

The classification of cyclic vectors in the first case is the following theorem.

Theorem 3. *Suppose $\dim(aH^2)^\perp < \infty$. Denote by $\lambda_1, \lambda_2, ..., \lambda_s$ the Carathéodory angular derivatives of b (that s is finite is a part of the theorem). Then*

(i) every function $h \in \mathcal{H}(b)$ extends non-tangentially to λ_j,

(ii) $f \in \mathcal{H}(b)$ is cyclic if and only if f is outer and $f(\lambda_j) \neq 0$, for all $j = 1, 2, ..., s$.

The second case is much harder and very little is known. In the paper several sufficient and some necessary conditions were given. Here we highlight an application which is definitive.

Theorem 4. *Let $\mathcal{H}(b)$ be a shift invariant de Branges-Rovnyak space.*

(i) For each $\lambda \in \mathbb{D}$ the kernel $k_\lambda^b(z) = (1 - \overline{b(\lambda)}b(z))/(1 - \overline{\lambda}z)$ is a cyclic vector.

(ii) The function b is cyclic if and only if it is outer.

There is one known case where $\dim(aH^2)^\perp = \infty$ and the cyclic vectors have a known simple description which is the case when $b = 2^{-1}(1 + \theta)$, where θ is an inner function. In this case $(aH^2)^\perp = K_\theta$ and the following theorem was obtained in paper I.

Theorem 5. *Let θ be a non-constant inner function and $b = (1 + \theta)/2$. Then*

(i) $\mathcal{H}(b) = \frac{1}{2}(1 - \theta)H^2 \oplus K_\theta$, where the orthogonal sum is taken with respect to the norm in $\mathcal{H}(b)$.

(ii) All functions in $\mathcal{H}(b)$ have non-tangential limits σ-a.e., where σ is the measure, such that

$$\frac{1 - |\theta|^2}{|1 - \theta|^2} = \int_{\mathbb{T}} \frac{1 - |z|^2}{|\zeta - z|^2} d\sigma(\zeta).$$

(iii) Let $f \in \mathcal{H}(b)$ be an outer function. Then f is cyclic if and only if f is non-zero σ-a.e.

Paper II

Paper II [7] is somewhat outside the main theme of this thesis although it shares some common features with paper IV. It concerns Chebyshev polynomials in the plane. For a fixed compact set $E \subset \mathbb{C}$ and positive integer n the Chebyshev polynomial T_n^E is the unique monic polynomial of degree n minimizing the supremum norm on E. The classical Chebyshev polynomials correspond to the set $E = [-1, 1]$. In paper II we consider the case when E is the unit circle and with the weight $w_s = (z - 1)^s$, $s > 0$. Our reason for considering this weight is that through a change of variables

this problem corresponds to the problem of determining the (unweighted) Chebyshev polynomials on lemniscates of the form

$$E_m = \{z \in \mathbb{C} : |z^m - 1| = 1\}.$$

Our main result is that $T_n^{E_m}$ have asymptotically minimal norm.

Theorem 6. *With E_m and $T_n^{E_m}$ as above we have*

$$\lim_{n\to\infty} \|T_n^{E_m}\|_{E_m} = 1.$$

To obtain this result we consider the same minimization problem with zeros constrained to the unit circle

$$\inf\left\{\sup_{z\in\mathbb{T}}|p(z)w_s(z)| : p \in \mathscr{P}_{n,\mathbb{T}}\right\},$$

where $\mathscr{P}_{n,\mathbb{T}}$ denotes the set of monic polynomials of degree at most n with zeros on $\mathbb{T}$. Let $\mathring{T}_n^{E_m}$ be a solution to the extremal problem (in fact we prove that it is unique). The Gauss-Lucas theorem tells us that the critical points of $\mathring{T}_n^{E_m}$ lie inside the closed unit disc. Surprisingly the derivative of $\mathring{T}_n^{E_m}$ is related to the unconstrained extremal, $T_n^{E_m}$, in a simple way.

Theorem 7. *With $\mathring{T}_n^{E_m}$ and $T_n^{E_m}$ as above we have*

$$\frac{1}{s+n+1}\frac{d}{dz}\left(w_{s+1}(z)\mathring{T}_n^{E_m}(z)\right) = w_s(z)T_n^{E_m}(z).$$

Using this theorem one can estimate the norm of $T_n^{E_m}$ by using a conformal map onto a compact real interval and a theorem of Bernstein on real Chebyshev polynomials. The main tool to prove theorem 7 is the following generalization of the classical Erdős-Lax inequality.

Theorem 8. *Let $k = 1, 2, ..., N$ and $s_k \geq 1$ be real numbers and $a_1, ..., a_N \in \mathbb{T}$. The analytic function*

$$f(z) = c\prod_{k=1}^{N}(z - a_k)^{s_k}, \quad \textit{for } z \in \mathbb{D},$$

satisfies

$$\|f'\|_\infty \leq \frac{\sum_{k=1}^N s_k}{2}\|f\|_\infty.$$

In the case that all the s_k are integers f is a polynomial and the result is the Erdős-Lax inequality.

Paper III

Let $f : \mathbb{D} \to \mathbb{D}$ denote a holomorphic self map of the unit disc. The function f is said to have an angular derivative at $\lambda \in \mathbb{T}$ in the sense of Carathéodory if the following two conditions hold

(i) f has a non-tangential limit at λ and $|f(\lambda)| = 1$.

(ii) The derivative, f', has finite non-tangential limit at λ.

The set of Carathéodory angular derivatives will be denoted by $C(f)$. In paper III we characterized $C(f)$ in the case that f is nonextreme, i.e.

$$\int_{\mathbb{T}} \log(1 - |f|)dm > -\infty. \tag{1.4}$$

The relevant notion is that of a Beurling-Carleson set. A closed set $E \subset \mathbb{T}$ of measure zero is called Beurling-Carleson if

$$-\sum |I| \log |I| < \infty,$$

where the sum is over the maximal complementary arcs of E. In paper III [4] we obtained

Theorem 9. *Let $f : \mathbb{D} \to \mathbb{D}$ be holomorphic. If f satisfies* (1.4)*, then $C(f)$ is a countable union of Beurling-Carleson sets. Conversely, if E is a countable union of Beurling-Carleson sets then there exists an $f : \mathbb{D} \to \mathbb{D}$, such that $C(f) = E$.*

We also obtained local variants of the theorem. Paper III is related to paper I since it turns out that every function in the de Branges-Rovnyak space $\mathcal{H}(f)$ extends radially to $\lambda \in \mathbb{T}$ if and only if $\lambda \in C(f)$.

Paper IV

A classical inequality due to Hörmander is the following.

Theorem 10 (Hörmander [14])**.** *Let f be an entire function of exponential type at most a which is real on the real axis. Suppose $|f(x)| \leq 1$ for all real x and $f(0) = 1$. Then*

$$f(x) \geq \cos(ax), \ \textit{for} \ -\frac{\pi}{a} \leq x \leq \frac{\pi}{a}.$$

In paper IV we generalize this inequality to the de Branges space setting. Let E be a Hermite-Biehler function with no real zeros. We write $E = A + iB$ where A and B are the real entire functions

$$A(z) = \frac{E(z) + E(z)^{\#}}{2}, \text{ and } B(z) = \frac{E(z) - E(z)^{\#}}{2i}.$$

The functions A and B play the role of cosine and sine in the case when E is an exponential function. The $p = \infty$ de Branges space is

$$\mathcal{H}^{\infty}(E) = \left\{ f \in \mathcal{O}(\mathbb{C}) : f/E, f^{\#}/E \in H^{\infty} \right\}.$$

The norm is given by $\|f\| = \|f/E\|_{\infty}$. Our main result in paper IV [5] was the next theorem.

Theorem 11. *Let E be a Hermite-Biehler function with no real zeros, $E(0) = 1$ and $f \in \mathcal{H}^{\infty}(E)$ be real entire. Suppose $f(0) = 1$. Then*

$$f(x) \geq A(x), \text{ for } b_{-1} \leq x \leq b_1,$$

where b_{-1} is the first zero of B to the left of the origin and b_1 is the first zero of B to the right of the origin.

Hörmander's inequality, which can be seen as a decay estimate at a global maximum, has a corresponding growth estimate at a zero which is due to Beurling.

Theorem 12 (Beurling [9])**.** *Let f be an entire function of exponential type at most a. Suppose $|f(x)| \leq 1$ for all real x and $f(0) = 0$. Then*

$$|f(x)| \leq |\sin(ax)|, \quad \frac{-\pi}{2a} \leq x \leq \frac{\pi}{2a}.$$

Beurling proved his result using his theorems on Balayage of Fourier-Stieltjes transforms. The method from paper IV also works to prove this theorem and yields the following generalization to the de Branges space setting. As I was unaware of this theorem when paper IV was written it does not appear there, but it seems noteworthy enough to mention here.

Theorem 13. *Let E be an HB function with no real zeros and $E(0) = 1$. Suppose $f \in \mathcal{H}^{\infty}(E)$ has a zero at the origin, $f(0) = 0$. Then*

$$|f(x)| \leq \|f/E\|_{\infty} |B(x)|, \quad a_{-1} \leq x \leq a_1,$$

where a_{-1} is the largest negative zero and a_1 is the smallest positive zero of A.

Paper V

Paper V [1] is possibly the most substantial in the thesis. The full program described in the introduction is carried out for the class of operators defined by a Krein-de Branges canonical system. However, first we shall consider a motivating example. Let V denote the Volterra operator

$$Vf(x) = \int_0^x f(t)dt,$$

D the derivative operator

$$Df(x) = f'(x),$$

and M_x multiplication by x

$$M_x f(x) = xf(x).$$

We are interested in studying closed invariant subspaces of the above operators. V and M_x act as bounded linear transformations on $L^2(0,1)$ and thus it is natural to consider them on this space. The derivative does not and we instead consider $C^\infty(\mathbb{R})$ with topology given by uniform convergence of all derivatives on compact subsets. An invariant subspace is then a closed subspace $L \subset X$ (where X is $L^2(0,1)$ or C^∞ as the case may be) such that $TL \subset L$ (where $T = V, D, M_x$). Because of the algebraic relationships

$$DV = I,$$

and

$$[D, M_x] = DM_x - M_x D = I,$$

the invariant subspaces of the three operators turn out to be related. The classification of the invariant subspaces of M_x and V are classical, but the invariant subspaces of D are still not completely understood. In [2] Aleman and Korenblum considered these invariant subspaces. Our work is a considerable generalization of the results obtained there.

Now we go back to Krein-de Branges canonical systems. Let $H : (0,\infty) \to \mathbb{R}^{2\times 2}$ be a positive, $H \geq 0$, real-matrix valued function on $(0,\infty)$. To simplify the exposition we assume that H has constant trace equal to 1. A Krein-de Branges canonical system is a differential equation of the form

$$\Omega \partial_t X(t) = \lambda H(t) X(t), \qquad \Omega = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \ \lambda \in \mathbb{C}. \tag{1.5}$$

The matrix-valued function H is called the *Hamiltonian* of the canonical system. Formally, equation (1.5) has the form of an eigenvalue problem for the differential operator

$$D = H(t)^{-1}\Omega\partial_t.$$

However, the Hamiltonian may be singular and it is important for the theory to allow for that possibility. It is possible to make sense of the operator and define a Hilbert space $\mathscr{H}$ where it acts as an unbounded operator. We then introduce a subspace of "smooth" vectors defined by

$$C^\infty(D) = \bigcap_{n\geq 0} \mathrm{Dom}(D^n).$$

The space $C^\infty(D)$ is then given a Frechet space structure such that D acts continuously and it makes sense to describe the closed subspaces $M \subset C^\infty(D)$ which are invariant under D. The invariant subspaces can be classified by their spectrum, σ_M, which is defined to be the spectrum of D restricted to M. There are three possibilities.

(i) $\sigma_M = \varnothing$,

(ii) σ_M is discrete and nonvoid,

(iii) $\sigma_M = \mathbb{C}$.

In the paper we consider subspaces of the first type, because they are the simplest and also must be understood before tackling the other cases. We give a complete description of such subspaces using a model defined by a Green's operator for D. The Green's operator in question is defined by

$$Vf(x) = -\Omega \int_0^x H(t)f(t)dt.$$

Which solves the equation $DVf = f$ on the appropriately defined Hilbert space $\mathscr{H}$. The model is then given by

$$Uf(z) = \langle f, (I - \overline{z}V)^{-1}\phi_0\rangle_{\mathscr{H}},$$

where ϕ_0 is a non-trivial element of the kernel of D. It turns out that the model

$$U\mathscr{H} = \{Uf : f \in \mathscr{H}\},$$

is nearly a de Branges space of entire function in that there exists a Hermite-Biehler function E with no real zeros and a real number a, such that

$$U\mathscr{H} = e^{iaz}\mathcal{H}(E).$$

Deep theorems from de Branges spaces are then used to classify the V invariant subspaces and D invariant subspaces with void spectrum. Since the statements are rather long we refer to the paper for them.

Paper VI

In paper VI [6] we go back to the central problem stated in the introduction of trying to understand which functions belong to a given model space. This time we consider the somewhat more general setup of Toeplitz kernels. Let $U \in L^\infty(\mathbb{R})$. The Toeplitz operator, $T_U : H^2 \to H^2$ with symbol U is defined by the formula $T_U f = P_+ U f$, where P_+ is the orthogonal projection from L^2 onto H^2. If θ is an inner function and $U = \overline{\theta}$, then $\mathrm{Ker}(T_{\overline{\theta}}) = K_\theta$. Here we shall consider only real analytic and unimodular symbols U. In that case all elements in the Toeplitz kernel turn out to be real analytic as well. We shall also consider more general L^p variants

$$N^p(U) = \left\{ f \in H^p : \overline{Uf} \in H^p \right\}, \ p \geq 1.$$

Let $d\mathbb{P} = (1+t^2)^{-1} dt$ denote the Poisson measure. To decide if a given nonnegative and p-summable function f satisfying $\log f \in L^1(\mathbb{P})$ is the absolute value of a function in $N^p(U)$ we shall use a lemma which goes back at least to Dyakonov [11]. We need the Hilbert transform which we define to be

$$\widetilde{f}(x) = \mathscr{H}(f)(x) = \frac{1}{\pi} \int_{-\infty}^{\infty} \left(\frac{1}{x-t} + \frac{t}{1+t^2} \right) f(t) dt.$$

where the integral is taken in the principal value sense.

Lemma 1. *Let f be a nonnegative function in L^p and $\log f \in L^1(\mathbb{P})$. Then f equals the absolute value of a function in $N^p(U)$ if and only if*

$$\widetilde{2\log f} + \arg(U) = -\arg(I) + 2\pi n,$$

for some meromorphic inner function I and measurable integer-valued function $n : \mathbb{R} \to \mathbb{Z}$.

Of course, in general it is very difficult to decide if the equation in the lemma has a solution. To get around this we use the following trick. Suppose m is a positive function which is bounded above and below. Then $u = fm$ also satisfies the hypothesis of the lemma and hence is the absolute value of a $N^p(U)$ function if and only if

$$\widetilde{2\log m} = -\widetilde{2\log f} - \arg(U) - \arg(I) + 2\pi n,$$

for some meromorphic inner function I and measurable integer-valued function $n : \mathbb{R} \to \mathbb{Z}$. Using that $\widetilde{\widetilde{m}} = -m + c$ we are lead to the equation

$$2\log m = \mathscr{H}\left(\widetilde{2\log f} + \arg(U) + \arg(I) + 2\pi n \right).$$

The left hand-side is just a bounded function. Thus if for a given f and U we can construct a meromorphic inner function and an integer valued function n, such that the function

$$2\widetilde{\log f} + \arg(U) + \arg(I) + 2\pi n$$

is the Hilbert transform of a bounded function then we will have proved that f is comparable to the absolute value of a $N^p(U)$ function. This is a more tractable, but still difficult, problem. In the paper we use some additional tricks to allow for slightly more growth. In paper VI we attempt to solve this equation with $n \equiv 0$. The conditions on f and U which are natural to put turn out to be that $2\widetilde{\log f} + \arg(U)$ is a decreasing (or almost decreasing) function. The argument of a meromorphic inner function is an increasing function and so the question becomes how well a general increasing function can be approximated by the argument of a meromorphic inner function. Thus the abstract situation becomes that of a minimizing $h = \alpha - \arg(I)$ where α is increasing and I is meromorphic inner. To obtain the desired estimates on the Hilbert transform we must control both the growth of h and its derivative. Our main result lets us do this in the case that α is what we call a *regular locally doubling weight* which for arguments of meromorphic inner functions is equivalent to the property that $d\mu = \alpha'(x)dx$ is a doubling measure on short intervals, i.e. there exists a constant $C > 0$, such that $\mu(2I) \leq C\mu(I)$, whenever $\mu(I) \leq 1$.

Theorem 14. *Let $\alpha : \mathbb{R} \to \mathbb{R}$ be a smooth regular locally doubling weight satisfying*

$$(1 + x^2)^{-\kappa/2} \lesssim \alpha'(x), \text{ for some } \kappa < 1.$$

Then there exists a meromorphic inner function J and a positive integer N_0, such that

$$|\alpha(x) - \arg J(x)| \leq 2\pi, \text{ for } x \in \mathbb{R},$$

and

$$|J'(x)| \lesssim \alpha'(x)^{N_0}(1 + x^2)^{N_0}, \text{ for } x \in \mathbb{R},$$

and

$$(1 + \lambda^2)^{-N_0}\alpha'(\lambda)^{-N_0} \lesssim |J'(\lambda)| \text{ for all } \lambda \in \Lambda,$$

where $\Lambda = \{\lambda \in \mathbb{R} : \alpha(\lambda) = 0 \mod 2\pi\}$.

Although, it is natural to assume that $\alpha(\pm\infty) = \pm\infty$ the lower bound on α' is a blemish. This result is then applied to the construction of functions with prescribed zeros and minorants in the Beurling-Malliavin sense.

Bibliography

[1] A. Aleman and A. Bergman. Invariant subspaces for generalized differentiation and Volterra operators. *arXiv preprint*, 2025.

[2] A. Aleman and B. Korenblum. Derivation-invariant subspaces of C^∞. *Comput. Methods Funct. Theory*, 8(1-2):493–512, 2008.

[3] A. Bergman. On cyclicity in de Branges-Rovnyak spaces. *Indiana Univ. Math. J.*, 73(4):1307–1329, 2024.

[4] A. Bergman. A sharp entropy condition for the density of angular derivatives. *C. R. Math. Acad. Sci. Paris*, 363:29–33, 2025.

[5] A. Bergman. Hörmander's inequality and point evaluations in de Branges space. *Rev. Mat. Iberoam.*, 42(2):551–572, 2026.

[6] A. Bergman. Inner approximations of doubling weights with applications to Beurling-Malliavin theory in Toeplitz kernels. *arXiv Preprint*, 2026.

[7] A. Bergman and O. Rubin. Chebyshev polynomials corresponding to a vanishing weight. *J. Approx. Theory*, 301:Paper No. 106048, 25, 2024.

[8] A. Beurling. On two problems concerning linear transformations in Hilbert space. *Acta Math.*, 81:239–255, 1948.

[9] A. Beurling. *The collected works of Arne Beurling. Vol. 2*. Contemporary Mathematicians. Birkhäuser Boston, Inc., Boston, MA, 1989. Harmonic analysis.

[10] P. L. Duren. *Theory of H^p spaces*, volume Vol. 38 of *Pure and Applied Mathematics*. Academic Press, New York-London, 1970.

[11] K. M. Dyakonov. Moduli and arguments of analytic functions from subspaces in H^p that are invariant under the backward shift operator. *Sibirsk. Mat. Zh.*, 31(6):64–79, 1990.

[12] S. R. Garcia and W. T. Ross. Model spaces: a survey. In *Invariant subspaces of the shift operator*, volume 638 of *Contemp. Math.*, pages 197–245. Amer. Math. Soc., Providence, RI, 2015.

[13] J. B. Garnett. *Bounded analytic functions*, volume 236 of *Graduate Texts in Mathematics*. Springer, New York, first edition, 2007.

[14] L. Hörmander. Some inequalities for functions of exponential type. *Math. Scand.*, 3:21–27, 1955.

[15] N. Makarov and A. Poltoratski. Meromorphic inner functions, Toeplitz kernels and the uncertainty principle. In *Perspectives in analysis*, volume 27 of *Math. Phys. Stud.*, pages 185–252. Springer, Berlin, 2005.

[16] B. Sz.-Nagy, C. Foias, H. Bercovici, and L. Kérchy. *Harmonic analysis of operators on Hilbert space*. Universitext. Springer, New York, second edition, 2010.

Scientific publications

Minor differences between previously published versions of the papers and the versions in this thesis may occur.

Author contributions

In paper II the main contributions were due to Olof Rubin. The only section in which I contributed substantially was section 2 on the generalization of the Erdős-Lax inequality.

In paper V both authors contributed equally in the formulation of the problems, proofs of theorems, and writing.

All other papers were solo authored.



Printed by Media-Tryck, Lund 2026 NORDIC SWAN ECOLABEL 3041 0903

LUND UNIVERSITY

Doctoral Theses in Mathematical Sciences 2026:4
ISBN 978-91-8104-931-2
ISSN 1404-0034